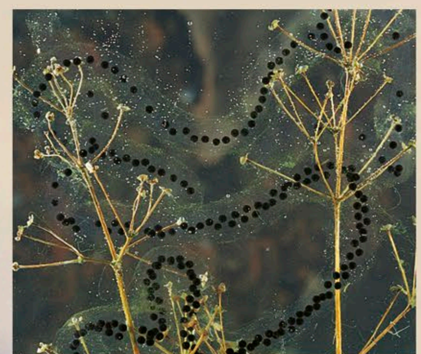
Reproduction

Once males and females have located one another (see facing page), mating can occur. The male adopts a position known as amplexus, in which he clasps the female from above (see right). Fertilization takes place while the pair are in amplexus, which may last for a period ranging from a few minutes to several days, depending on the species. In all but a small number of species, fertilization is external, the male shedding sperm onto the eggs as they emerge from the female. Each species has a characteristic pattern of egg production and uses a specific type of site for egg deposition. Eggs may be produced singly, in clumps, or in strings. The female usually places them in water, because their development can occur only in moist conditions. Eggs may be dispersed into the water, where they may float or sink, they may be wrapped around vegetation, or they may be glued to plants, logs, or rocks.

AMPLEXUS
Amplexus may take place on land or in water—these tree frogs (right) mate in the branches of trees. Depending on the species, the male may clasp the female just behind her front limbs or just in front of her back limbs. In many species, males outnumber females in breeding groups, and males in amplexus are often attacked by rivals seeking to displace them.



EGGS
The number of eggs produced by a female varies greatly among species, from less than 20 to many thousand. The eggs of the European common toad (left), which are suspended in long strings of jelly, are wrapped around the underwater stems of aquatic plants.

Defense

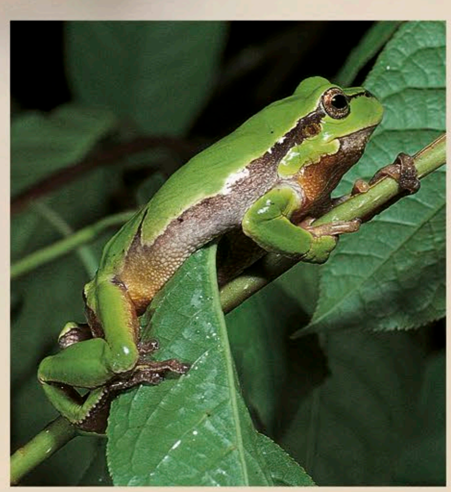
Frogs and toads typically lack weapons, such as large teeth or claws, that they can use to defend themselves. Most rely on being able to escape by jumping, usually toward open water. Many produce secretions from glands in their skin that make them distasteful or poisonous, sometimes lethally so. Others rely on being well camouflaged. In general, tadpoles lack defenses and are eaten by various predators, notably fish and the larvae of insects, such as dragonflies. Being produced in large numbers means that at least a few survive.

WARNING COLORATION
The poison-dart frog (left) has had to evolve highly poisonous skin secretions because many of its predators (including snakes and spiders) have evolved a resistance to milder toxins. Many species that are distasteful or toxic are vividly colored, which acts as a warning to predators.

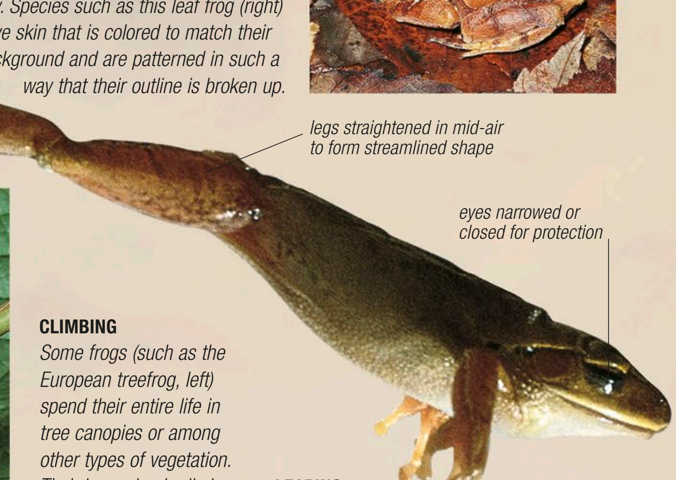
CAMOUFLAGE
Most frogs and toads are active only at night and remain entirely motionless by day. Species such as this leaf frog (right) have skin that is colored to match their background and are patterned in such a way that their outline is broken up.

Movement

Frogs and toads move in a wide variety of ways. Most toads are ground dwellers and walk, run, or hop over the ground; few are able to leap large distances. Leaping is rarely used for moving from place to place but is primarily a means of escape from predators. Not all frogs and toads jump: many have back limbs adapted for other types of movement, such as swimming, climbing, and (in a few species) gliding. Burrowing is important for many species, enabling them to hide during the day, or for many weeks or months when it is too cold, hot, or dry for them to be active. They may simply push their way under rocks or logs, or into piles of plant debris, or they may be able to dig deep into the ground. Some desert species can survive being buried for up to 2 or 3 years.



CLIMBING
Some frogs (such as the European treefrog, left) spend their entire life in tree canopies or among other types of vegetation. Their long, slender limbs enable them to jump or straddle gaps between branches, while the adhesive disks on their digits allow them to climb vertical surfaces.



LEAPING
Frogs and toads launch themselves into the air with an explosive burst of energy from their long hind legs. On landing, the front legs are pushed forward to cushion the impact.